RENT CALCULATOR PROJECT

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1. Introduction

The **Shared Expenses Calculator** is a simple utility program designed to **automate the fair division of common household expenses**, such as rent, food, and electricity, among a group of people living together (e.g., in a hostel, room, or apartment). The primary goal of this project is to calculate the **individual contribution** each person must pay to cover the total shared expenditure. This eliminates the need for manual calculations, ensuring accuracy and fairness in expense splitting.

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2. Working Principle and Methodology

The program's methodology is based on four main steps: **Input Collection**, **Calculation of Total Bill**, **Division of Expenses**, and **Output Generation**.

A. Input Collection

The program starts by prompting the user to enter five specific values, which are converted to **integers ()** for mathematical operations:

- rent: The total monthly rent for the shared accommodation.
- food: The total Charges of Mess.
- electricity_spend: The total number of electricity units consumed.
- charge_per_unit: The monetary cost for a single electricity unit.
- persons: The total number of individuals sharing the expenses.

B. Calculation of Total Bill

The total cost of electricity is calculated first by multiplying the units consumed by the rate per unit:

This calculated electricity cost is then added to the fixed rent and food expenses to determine the **Grand Total Shared Expense**.

C. Division of Expenses

The Grand Total Shared Expense is then divided equally among the number of persons. The division uses the **floor division operator (//)** to ensure the result (output) is a **whole number (integer)**, representing the amount each person must pay.

$$output = (food + rent + total_bill) // persons$$

D. Output Generation

Finally, the program prints the calculated amount (output) with a clear message, informing the user of **"Each person will pay = "** the calculated amount.

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That's a great simple project! Here is a comprehensive project report based on the Python code you provided for a **Rent and Shared Expenses Calculator**.

3. ✅ Conclusion

The **Shared Expenses Calculator** successfully fulfills its objective by efficiently calculating the equal share of common household expenses for each individual. The program is **straightforward, easy to use, and highly effective** for a quick and fair division of costs. It demonstrates fundamental Python concepts including user input, variable assignment, basic arithmetic operations, and formatted output.

The project can be further enhanced by incorporating features like **handling variable expenses** (e.g., individual contributions to food), **using floating-point numbers** for precise currency calculations, and **error handling** (e.g., ensuring the number of persons is not zero).

Would you like me to modify the code to use **standard division (/)** for more precise monetary calculations, or perhaps add a feature for **error handling**?

program.py > ...

```python
## Inupts we need from the user
# Total rent
# Total food ordered for snacking
# Electricity units spend
# Charge per unit
# Persons living in room/flat

## Output
# Total amount you've to pay is

rent = int(input("Enter your hostel/flat rent = "))
food = int(input("Enter the charges of mess = "))
electricity_spend = int(input("Enter the total of electricity spend = "))
charge_per_unit = int(input("Enter the charge per unit = "))
persons = int(input("Enter the number of persons living in room/flat = "))

total_bill = electricity_spend * charge_per_unit

output = (food + rent + total_bill) // persons

print("Each person will pay = ", output)
```